



Suggested FYP Topics

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Expectations

- Suppose to have mild to strong background on maths and/or programming
 - Good problem-solving skills
 - Suppose to find out novel results
 - Somehow research-based: either theoretical or practical
 - Good-enough results may be published to enhance your CV
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Outline

- Network Coding over Multipath Networks
- Optimizations in Batched Network Coding
- Experiments in Steganography and Information Hiding
- Self-Proposed Topics

Network Coding



- Packet loss? No problem (?)
- Not just store-and-forward



HFY1: Network Coding over Multipath Networks

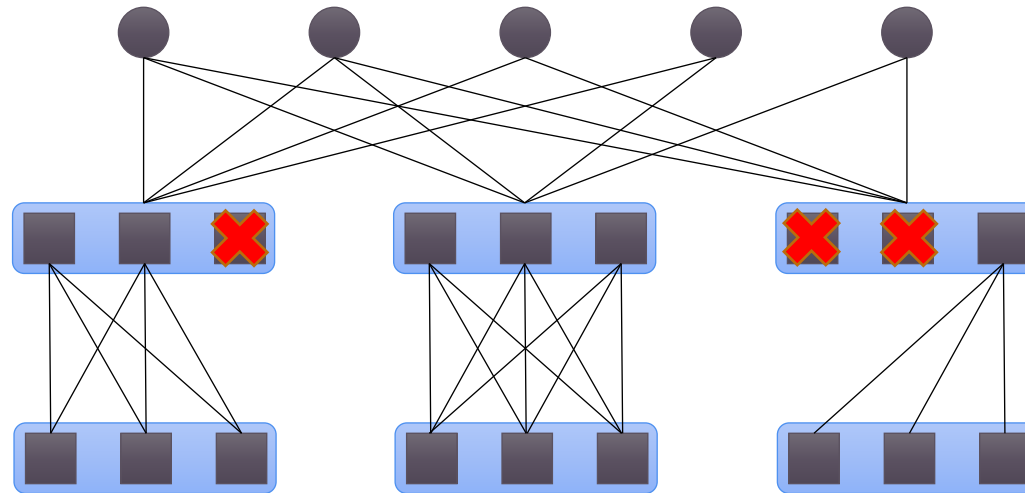
- Traditional networking: Use a single path
 - Why don't use the other paths?
 - How to divide the data and regroup the data later?
 - Network coding is an approach
 - What is still missing?
- 



HFY1: Network Coding over Multipath Networks

- Find the available paths
 - Need to use/design a protocol
 - Control the transmission rates via each path
 - Design a scheme
 - Protocol Design
 - Design a protocol for achieving the goal
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HFY2: Optimizations in Batched Network Coding



- A variant of network coding: Recoding are restricted within the same batch

HFY2: Optimizations in Batched Network Coding

- Open problems:
 - When to recode?
 - Too late = high latency. Too early = bad throughput
 - How to schedule the transmission?
 - Interleaving? Reduce delay?
 - How many recoded packets?
 - Especially when the channel condition is changing
 - How to enhance security?
 - How to balance header overhead and throughput?

HFY2: Optimizations in Batched Network Coding

- No limitation on the techniques
 - Machine learning? OK
 - Maths analysis? OK
 - Simulation? OK

HFY3: Experiments in Steganography and Information Hiding

- Steganography: The art of hiding secret messages
 - Hide in images, audio, etc.
 - Hide in channel noise, ciphertext, QR codes, etc.
- Design / Implementation
 - Using image library, crypto library, etc.
- Steganalysis
 - Machine learning / statistics
- What you need to learn: ECC, statistical analysis, etc.

HFY4: Self-Proposed Topics

- Anything you want to do!
 - Implement “useful” applications
 - Investigate interesting problems
 - Conduct research on your own topic
- Practical or theory? Your choice!
 - Suppose to give novel results
- Preferred fields: computer networking, Internet of Things, data compression, operating systems, network coding, communication, optimization

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Thank you